

SEAN GRATE

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EDUCATION

Auburn University, Auburn

August 2020 –August 2025

Ph.D., Mathematics

Advisor: Hal Schenck

Dissertation Title: *Problems in Computational Algebraic Geometry: Lefschetz Properties and Toric Varieties*

University of Kentucky, Lexington

August 2016 –May 2020

B.S. in Mathematics with minors in History and Classics, *cum laude*

PROFESSIONAL EXPERIENCE

Iowa State University Department of Mathematics

August 2025–Present

Postdoctoral Research Associate

Auburn University Department of Mathematics and Statistics

August 2020–August 2025

Graduate Teaching Assistant

University of Kentucky Department of Computer Science

May 2019–August 2020

Undergraduate Research Assistant

PUBLICATIONS

- [5] Haile Gilroy, Melinda Lanius, and Sean Grate. “Graph theoretic reflection to foster alignment in coordinated courses”. *International Journal of Mathematical Education in Science and Technology* 0.0 (2025), pp. 1–18. DOI: 10.1080/0020739X.2025.2553315. eprint: <https://doi.org/10.1080/0020739X.2025.2553315>. URL: <https://doi.org/10.1080/0020739X.2025.2553315>.
- [4] Ayah Almousa, Sean Grate, Daoji Huang, Patricia Klein, Adam LaClair, Yuyuan Luo, and Joseph McDonough. “The MatrixSchubert package for Macaulay2”. *J. Softw. Algebra Geom.* 15.1 (2025), pp. 41–55. ISSN: 1948-7916. DOI: 10.2140/jsag.2025.15.41. URL: <https://doi.org/10.2140/jsag.2025.15.41>.
- [3] Nasrin Altafi, Roberta Di Gennaro, Federico Galetto, Sean Grate, Rosa M. Miró-Roig, Uwe Nagel, Alexandra Seceleanu, and Junzo Watanabe. “Betti numbers for connected sums of graded Gorenstein Artinian algebras”. *Trans. Amer. Math. Soc.* 378.2 (2025), pp. 1055–1080. ISSN: 0002-9947,1088-6850. DOI: 10.1090/tran/9286. URL: <https://doi.org/10.1090/tran/9286>.
- [2] Sean Grate and Hal Schenck. “Betti tables forcing failure of the weak Lefschetz property”. *Lefschetz properties—current and new directions*. Vol. 59. Springer INdAM Ser. Springer, Singapore, [2024] ©2024, pp. 155–164. ISBN: 978-981-97-3885-4; 978-981-97-3886-1. DOI: 10.1007/978-981-97-3886-1_8. URL: https://doi.org/10.1007/978-981-97-3886-1_8.
- [1] Hunter Blanton, Sean Grate, and Nathan Jacobs. “Surface Modeling for Airborne Lidar”. *IGARSS 2020 - 2020 IEEE International Geoscience and Remote Sensing Symposium*. 2020, pp. 1110–1113. DOI: 10.1109/IGARSS39084.2020.9323522.

PREPRINTS

- [3] Herman Chau, Spencer Daugherty, Sean Grate, J. Carlos Martínez Mori, and Anna Pun. “The combinatorics of stable Tamari posets”. In preparation.
- [2] Sean Grate and Jason McCullough. “Gorenstein graded Möbius algebras are Koszul”. In preparation.
- [1] Barbara Betti, Sean Grate, Thiago Holleben, and Flavio Salizzoni. *Incidence toric ideals and three-point functions*. 2026. arXiv: 2605.23849 [math.AC]. URL: <https://arxiv.org/abs/2605.23849>.

AWARDS AND HONORS

- Auburn University DMS Baskervill Endowed Mathematics Fellowship 2024
- Auburn University DMS Teaching Citation Award 2022, 2024
- Auburn University DMS Bennett Fellowship 2023
- Auburn University DMS Research Citation Award 2021, 2022
- Auburn University COSAM Outstanding GTA Award 2022
- Best Presentation at UK Computer Science Summer Research Program August 9th, 2019
- University of Kentucky Dean’s List Multiple semesters 2016-2020
- Kentucky Educational Excellence Scholarship (KEES) (\$2,225 per year) 2016-2020
- University of Kentucky Provost Scholarship (\$1,500 per year) 2016-2020

CONFERENCE TALKS

- TBD, Upcoming Researchers in Commutative Algebra (URiCA), August 2026
- “Gorenstein graded Möbius algebras are Koszul”, AMS Fall Spring Sectional, April 2026
- “Combinatorial bounds on the Castelnuovo-Mumford regularity of toric surfaces”, AMS Fall Central Sectional, October 2025
- “Combinatorial bounds on the Castelnuovo-Mumford regularity of toric surfaces”, Mathematical Conference of the Americas, July 2025
- “Castelnuovo-Mumford regularity of toric surfaces”, SIAM Conference on Applied Algebraic Geometry, July 2025
- “Combinatorial bounds on the Castelnuovo-Mumford regularity of toric surfaces”, AWM Research Symposium, May 2025
- “Castelnuovo-Mumford regularity of toric surfaces”, Joint Mathematics Meeting, January 2025
- “Castelnuovo-Mumford regularity of toric surfaces”, AMS Fall Central Sectional, September 2024
- “Betti tables forcing failure of the Weak Lefschetz Property”, Combinatorial Algebra meets Algebraic Combinatorics (CAAC), January 2024
- “Betti tables forcing failure of the Weak Lefschetz Property”, Workshop on Lefschetz Properties in Algebra, Geometry, Topology and Combinatorics, May 2023

SEMINAR TALKS

- “SVD is all you need”, Undergraduate Seminar, Grinnell College, May 2026
- “Combinatorial bounds on the Castelnuovo-Mumford regularity of toric surfaces”, Virtual, Syzygies and Mirror Symmetry Seminar, February 2025
- “Betti tables forcing failure of the weak Lefschetz property”, University of Nebraska-Lincoln, Commutative Algebra Seminar, October 2024
- “Betti tables and Lefschetz properties”, University of Nebraska-Lincoln, Commutative Algebra Reading Seminar, October 2024
- “Betti numbers of connected sums of graded Artinian Gorenstein algebras”, Purdue University, Commutative Algebra Seminar, October 2024
- “Betti tables forcing failure of the weak Lefschetz property”, Purdue University, Student Commutative Algebra Seminar, October 2024
- “Betti tables and Lefschetz properties”, University of Kentucky Algebra Seminar, November 2023
- “Lefschetz properties and Artinian rings”, McNeese State University Math Seminar, November 2023
- Iowa State University
 - “Gorenstein graded Möbius algebras are Koszul”, Algebra and Geometry Seminar, April 2026
 - “The stable Tamari lattice”, Discrete Math Seminar, November 2025
 - “Betti tables forcing failure of the weak Lefschetz property”, Algebra and Geometry Seminar, October 2025
- Auburn University
 - “The stable Tamari lattice”, Discrete Math Seminar, November 2024
 - “Betti numbers of connected sums of graded Artinian Gorenstein algebras”, Algebra Seminar, September 2024
 - “How-To: Making a website”, Graduate Student Seminar, August 2024
 - “Betti tables and Lefschetz properties”, Algebra Seminar, February 2024
 - “Suturing the severed didactic tetrahedron: Graph-Theoretic reflection to foster alignment in coordinated courses”, Auburn University DBER Seminar, February 2024
 - “Leveraging software for mathematics and graduate school”, Graduate Student Seminar, September 2023
 - “A brief introduction to tropical geometry”, Graduate Student Seminar, August 2022
 - “An overview of topological data analysis”, Math Club, February 2022
 - “A brief introduction to tropical geometry”, Algebra Seminar, November 2021
 - “Computations in topological data analysis”, Graduate Algebra Seminar, August 2021
 - “Tropical algebra”, Graduate Algebra Seminar, July 2021
 - “Geometry in noncommutative algebra”, First Year Graduate Student Seminar, January 2021
 - “What/Why/How of neural networks”, First Year Graduate Student Seminar, November 2020

POSTER PRESENTATIONS

- “Castelnuovo-Mumford regularity of toric surfaces”, Commutative Algebra in the South (CATS), April 2025
- “Castelnuovo-Mumford regularity of toric surfaces”, Mid-Atlantic Algebra, Geometry, and Combinatorics (MAAGC) Workshop, March 2025
- “Castelnuovo-Mumford regularity of toric surfaces”, Combinatorial Algebra meets Algebraic Combinatorics (CAAC), January 2025
- “Betti numbers of connected sums of graded Artinian Gorenstein algebras”, UweFest, August 2024
- “Betti tables forcing failure of the weak Lefschetz property”, BrianFest, August 2023

CONFERENCE AND WORKSHOP ORGANIZATION

- Co-organizer (with Michael Brown and John Cobb)
Macaulay2 Workshop
Auburn, AL February 2027
- Co-organizer (with Thiago Holleben and Anton Dochtermann)
AMS Special Session on algebraic combinatorics and combinatorial commutative algebra
New Orleans, LA October 2025

CONFERENCES AND WORKSHOPS ATTENDED

- Macaulay2 Workshop February 2027
- Weak and Strong Lefschetz Properties across Mathematics October 2026
- Upcoming Researchers in Commutative Algebra (URiCA) August 2026
- Building the Future of Macaulay2 May 2026
- Shaping the Future of Macaulay2 May 2026
- Kentucky-Ohio Algebra Alliance (KOALA) May 2026
- BIRS: Regularity in Algebra, Combinatorics, and Geometry April 2026
- AMS Spring Central Sectional Meeting April 2026
- CA+ March 2026
- Arizona-New Mexico Symposium on Commutative Algebra
and its Interactions: Geometric Combinatorics December 2025
- KUMUNU 2025 October 2025
- AMS Fall Central Sectional Meeting October 2025
- AMS Fall Southeastern Sectional Meeting October 2025
- Mathematical Congress of the Americas July 2025
- SIAM Conference on Applied Algebraic Geometry July 2025
- Collaborate@ICERM June 2025
- Workshop on Weak and Strong Lefschetz Properties Across Mathematics June 2025
- Workshop on the Applications of Commutative Algebra May 2025
- AWM Research Symposium May 2025

- Meetings in Applied Algebraic Geometry (MAAG) April 2025
- Macaulay2 Workshop April 2025
- Commutative Algebra in the South (CATS) April 2025
- Mid-Atlantic Algebra, Geometry, Combinatorics (MAAGC) March 2025
- CA+ March 2025
- Combinatorial Algebra meets Algebraic Combinatorics (CAAC) January 2025
- Apprenticeship Program in Commutative Algebra January 2025
- Joint Mathematics Meetings January 2025
- AMS Fall Central Sectional Meeting September 2024
- UweFest August 2024
- AMS MRC on Algebraic Combinatorics June 2024
- Computational Algebraic Geometry and String Theory June 2024
- JM Invariant at 60 May 2024
- Combinatorial Algebra meets Algebraic Combinatorics (CAAC) January 2024
- SIAM Texas-Lousiana Sectional Meeting November 2023
- BrianFest August 2023
- Macaulay2 Week June 2023
- SLMath Commutative Algebra Summer School May 2023
- Workshop on Lefschetz Properties in Algebra, Geometry, Topology and Combinatorics May 2023
- Commutative Algebra in the South (CATS) April 2023
- SLMath Tropical Geometry Summer School August 2022

TEACHING EXPERIENCE

- Iowa State University
 - Recitation
 - * MATH 1660 Calculus II Spring 2027, Fall 2025
 - Instructor of Record
 - * MATH 1500 Discrete Mathematics for Business and Social Sciences Fall 2026
 - * MATH 2070 Matrices and Linear Algebra Spring 2026
- Auburn University
 - Instructor of Record
 - * MATH 1610 Calculus I Spring 2022
 - * MATH 1120 Precalculus Algebra Fall 2021
 - Recitation Leader
 - * MATH 1610 Calculus I Fall 2022, Spring 2023, Fall 2023
 - * MATH 1620 Calculus II Spring 2024, Fall 2024

- Tutor
 - * MATH 1120 Precalculus Algebra
 - * MATH 1610 Calculus I
 - * MATH 1620 Calculus II
 - * MATH 1680 Business Calculus I

SERVICE

- Refereeing
 - Combinatorial Theory
 - Journal of Algebra
 - Journal of Software for Algebra and Geometry (technical reviewer)
- President of the Mathematics Club at Auburn University 2021-2024
- Secretary for the Auburn University DMS Graduate Student Council August 2022-August 2023
- Graduate Student Representative on the Auburn University DMS Graduate Student Council August 2021-August 2023

OUTREACH

- Destination STEM October 2022, October 2023
- Auburn Mathematical Puzzle Challenge (AMP'd) November 2022
- University of Kentucky Math Club 2016-2020
- Expanding Your Horizons workshop mentor 2019
- Julia Robinson Math Festival volunteer 2019

TECHNICAL STRENGTHS

Software & Tools	Python, Macaulay2, PyTorch, SageMath
Macaulay2 Packages	MatrixSchubert, Permutations